Firewall Project

3조

한준영, 서승덕, 백우진, 양혜원

목차

01 | 방화벽구성도

02 | 스위치 설정

03 | 라우터 설정

04 | 방화벽 1 설정

05 | 방화벽 2 설정

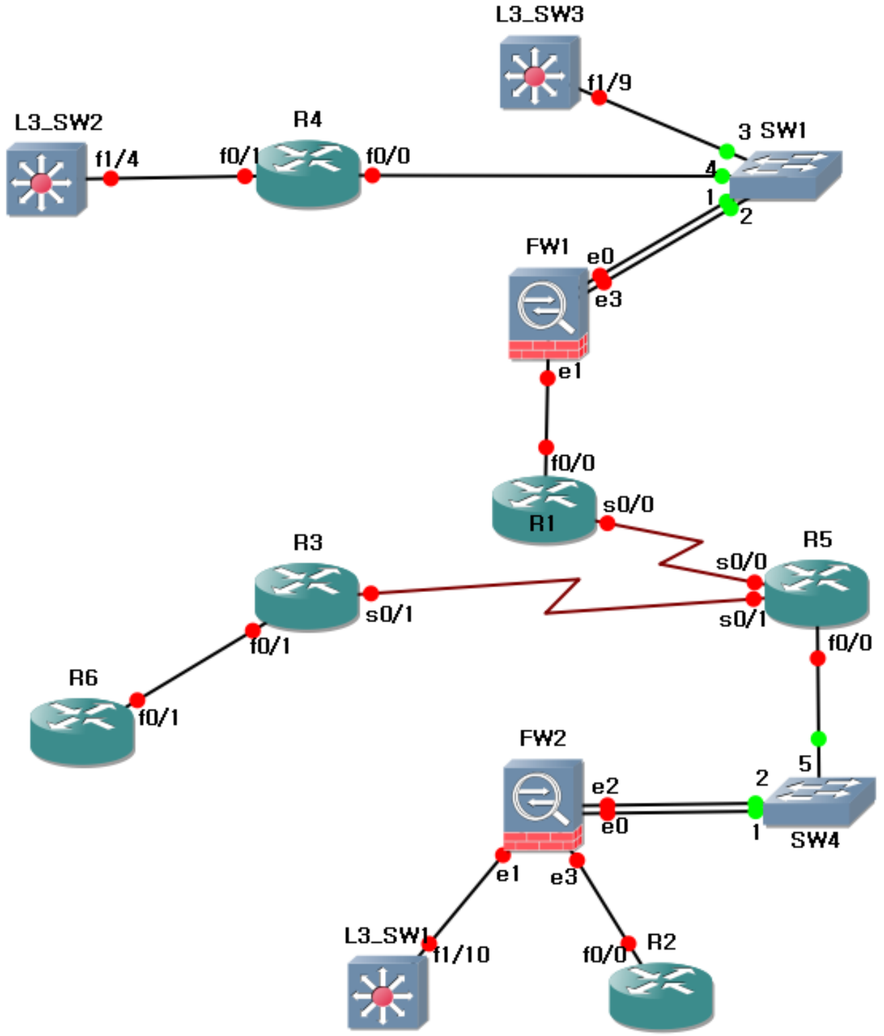
01

방화벽 구성도

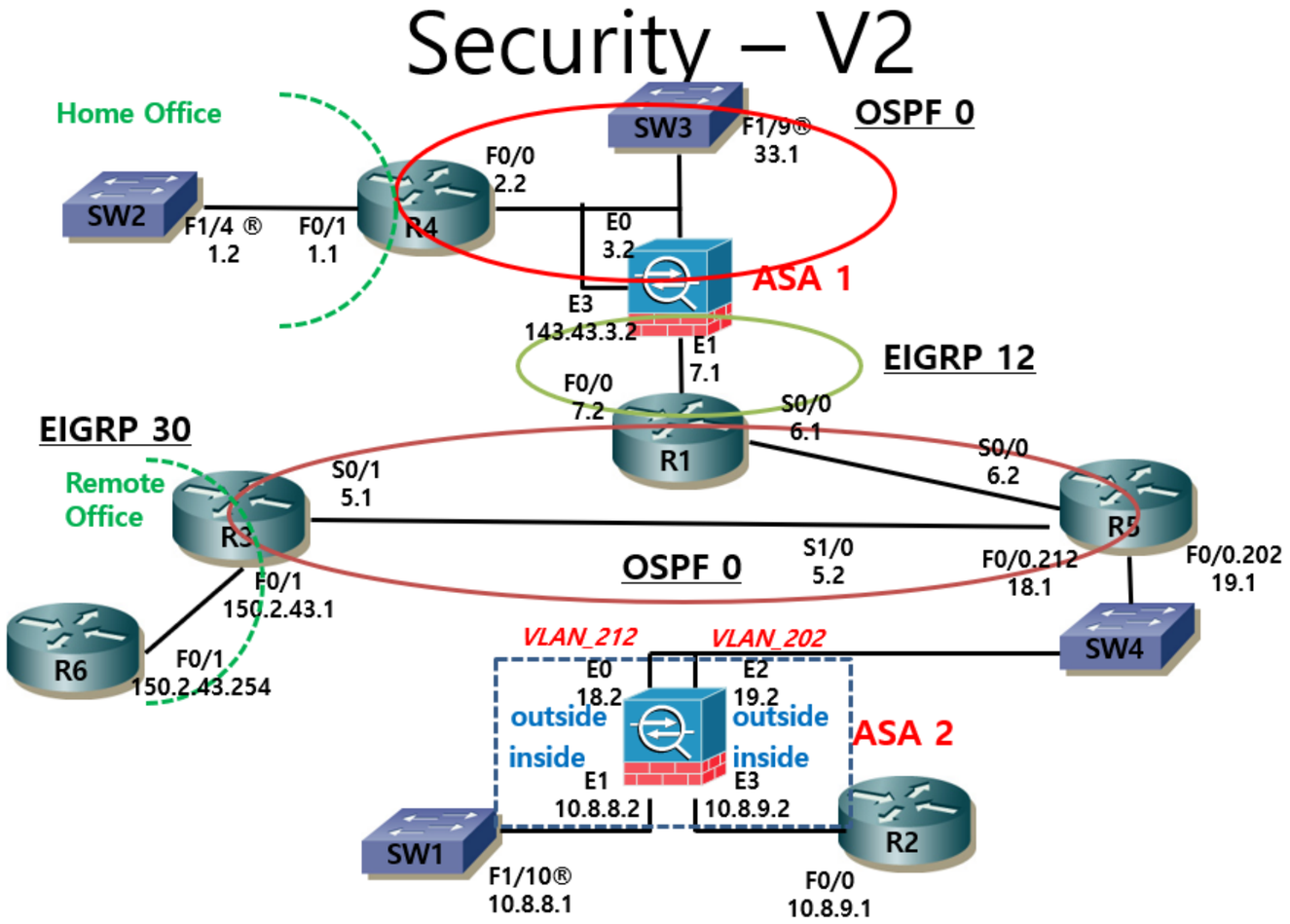
1-1 물리적 구성도

1-2 논리적 구성도

1-1 물리적 구성도


L3_SW3
f1/9
L3_SW2
f1/4
R4
f0/1
f0/0
3 SW1
4
1
2
FW1
e0
e3
e1
f0/0
R1
s0/0
R3
s0/1
f0/1
s0/0
s0/1
R5
f0/0
R6
f0/1
FW2
e2
e0
e1
e3
2
5
1
SW4
L3_SW1
f1/10
f0/0
R2

1-2 논리적 구성도


Security – V2
Home Office
SW3
OSPF 0
F1/9®
33.1
F0/0
2.2
SW2
F1/4 ®
1.2
F0/1
1.1
R4
E0
3.2
ASA 1
E3
143.43.3.2
E1
7.1
EIGRP 12
F0/0
7.2
S0/0
6.1
EIGRP 30
R1
S0/0
6.2
S0/1
5.1
Remote
Office
R3
R5
S1/0
5.2
OSPF 0
F0/0.212
18.1
F0/0.202
19.1
F0/1
150.2.43.1
R6
F0/1
150.2.43.254
SW4
VLAN_212
VLAN_202
E0
18.2
E2
19.2
outside
inside
outside
inside
ASA 2
E1
10.8.8.2
E3
10.8.9.2
SW1
R2
F1/10®
10.8.8.1
F0/0
10.8.9.1

02

스위치 설정

1-1 SW1

1-2 SW2

1-3 SW3

1-4 SW4

1-1 SW1

int f1/10
no sw
ip add 10.8.8.1 255.255.255.0

ip route 0.0.0.0 0.0.0.0 10.8.8.2

```
SW1#sh ip ro
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.8.8.2 to network 0.0.0.0

     10.0.0.0/24 is subnetted, 1 subnets
C       10.8.8.0 is directly connected, FastEthernet1/10
S*    0.0.0.0/0 [1/0] via 10.8.8.2
```

1-2 SW2

int f1/4
no sw
ip add 12.12.1.2 255.255.255.0

ip route 0.0.0.0 0.0.0.0 12.12.1.1

```
SW2#sh ip ro
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 12.12.11.1 to network 0.0.0.0

     12.0.0.0/24 is subnetted, 1 subnets
C       12.12.11.0 is directly connected, FastEthernet1/4
S*    0.0.0.0/0 [1/0] via 12.12.11.1
```

1-3 SW3

int f1/9
no sw
ip add 12.12.33.1 255.255.255.0

router ospf 1
router-id 9.9.9.9
net 12.12.33.1 0.0.0.0 a 0

```
L3_SW3#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     43.0.0.0/24 is subnetted, 10 subnets
O       12.12.1.0 [110/11] via 12.12.2.2, 00:31:09, FastEthernet1/10
O E2    12.12.5.0 [110/20] via 12.12.2.3, 00:22:54, FastEthernet1/10
O E2    12.12.6.0 [110/20] via 12.12.2.3, 00:22:54, FastEthernet1/10
O E2    12.12.7.0 [110/20] via 12.12.2.3, 00:22:54, FastEthernet1/10
O E2    12.12.11.0 [110/20] via 12.12.2.3, 00:22:54, FastEthernet1/10
O E2    12.12.33.0 [110/20] via 12.12.2.3, 00:19:59, FastEthernet1/10
C       12.12.2.0 is directly connected, FastEthernet1/10
O       12.12.44.0 [110/2] via 12.12.2.2, 00:19:41, FastEthernet1/10
O E2    12.12.55.0 [110/20] via 12.12.2.3, 00:20:11, FastEthernet1/10
O E2    12.12.66.0 [110/20] via 12.12.2.3, 00:22:56, FastEthernet1/10
     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O E2    12.8.2.2/32 [110/20] via 12.12.2.3, 00:11:44, FastEthernet1/10
O E2    12.8.8.0/24 [110/20] via 12.12.2.3, 00:11:46, FastEthernet1/10
O E2    12.8.9.0/24 [110/20] via 12.12.2.3, 00:11:46, FastEthernet1/10
     150.2.0.0/24 is subnetted, 1 subnets
O E2    150.2.43.0 [110/20] via 12.12.2.3, 00:22:57, FastEthernet1/10
```

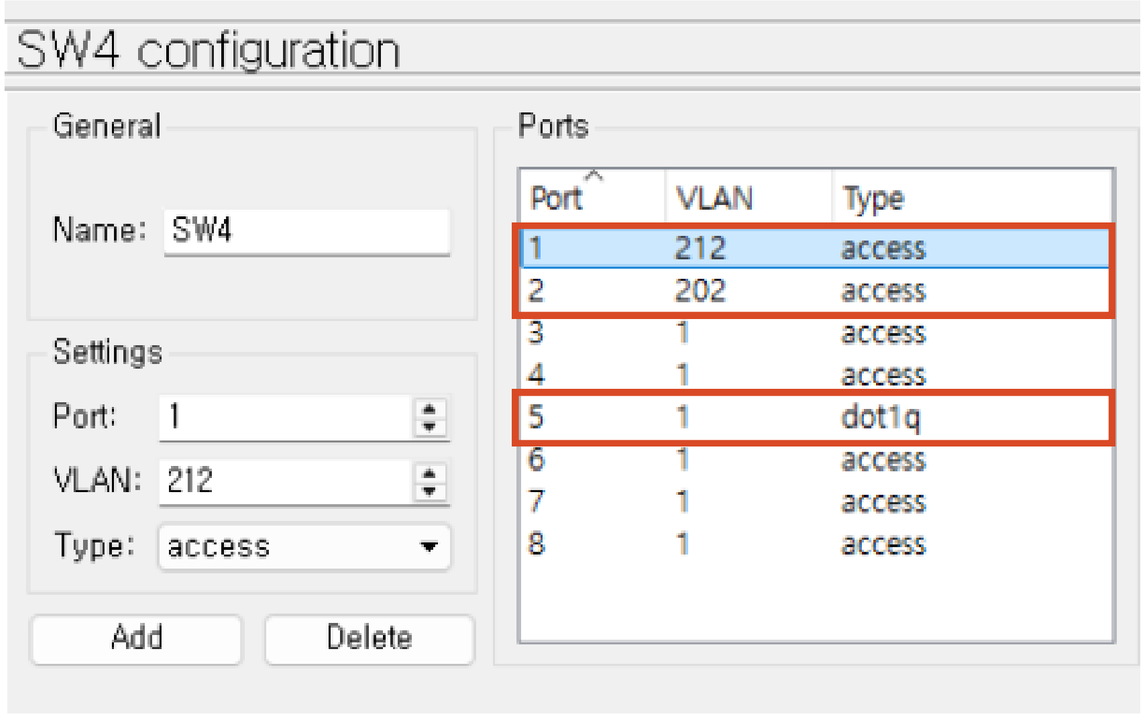
1-4 SW4

vlan 212
vlan 202

5번 포트
switchport trunk wncapsulation dot1q
switchport mode trunk

1번 포트
switchport mode access
switchport access vlan 212

2번 포트
switchport mode access
switchport access vlan 202



03

라우터 설정

1-1	R1
1-2	R2
1-3	R3
1-4	R4
1-5	R5
1-6	R6

1-1 R1

```
int lo0
ip add 12.12.11.1 255.255.255.0

int s0/0
no sh
ip add 12.12.6.1 255.255.255.0

int f0/0
no sh
ip add 12.12.7.2 255.255.255.0
```

```
router os 1
router-id 1.1.1.1
net 12.12.6.1 0.0.0.0 a 0
net 12.12.11.1 0.0.0.0 a 0
default-inf ori al
redi static sub
```

```
router eigrp 12
no auto
net 12.12.7.2 0.0.0.0
redi os 1 met 1 1 1 1 1
```

```
ip route 12.12.18.0 255.255.255.0 12.12.6.2
ip route 12.12.19.0 255.255.255.0 12.12.6.2
```

```
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     12.0.0.0/24 is subnetted, 10 subnets
D EX    12.12.1.0 [170/2560025856] via 12.12.7.1, 00:24:50, FastEthernet0/0
O       12.12.5.0 [110/128] via 12.12.6.2, 00:32:53, Serial0/0
C       12.12.6.0 is directly connected, Serial0/0
C       12.12.7.0 is directly connected, FastEthernet0/0
C       12.12.11.0 is directly connected, Loopback0
D       12.12.33.0 [110/129] via 12.12.6.2, 00:22:05, Serial0/0
D EX    12.12.34.0 [170/2560025856] via 12.12.7.1, 00:24:54, FastEthernet0/0
D EX    12.12.44.0 [170/2560025856] via 12.12.7.1, 00:21:46, FastEthernet0/0
O       12.12.55.0 [110/65] via 12.12.6.2, 00:22:17, Serial0/0
O E2    12.12.66.0 [110/20] via 12.12.6.2, 00:32:54, Serial0/0
     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O E2    10.8.2.2/32 [110/20] via 12.12.6.2, 00:13:50, Serial0/0
O E2    10.8.8.0/24 [110/20] via 12.12.6.2, 00:13:53, Serial0/0
O E2    10.8.9.0/24 [110/20] via 12.12.6.2, 00:13:53, Serial0/0
     150.2.0.0/24 is subnetted, 1 subnets
O E2    150.2.43.0 [110/20] via 12.12.6.2, 00:32:57, Serial0/0
```

1-2 R2

int lo0
ip add 10.8.2.2 255.255.255.0

int f0/0
ip add 10.8.9.1 255.255.255.0
no sh

ip route 0.0.0.0 0.0.0.0 10.8.9.2

```
R2#sh ip ro
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.8.9.2 to network 0.0.0.0

     10.0.0.0/24 is subnetted, 2 subnets
C       10.8.2.0 is directly connected, Loopback0
C       10.8.9.0 is directly connected, FastEthernet0/0
S*   0.0.0.0/0 [1/0] via 10.8.9.2
```

1-3 R3

int lo0
ip add 12.12.33.3 255.255.255.0

int s0/1
no sh
ip add 12.12.5.1 255.255.255.0

int f0/1
no sh
ip add 150.2.43.1 255.255.255.0

router os 1
router-id 3.3.3.3
net 12.12.33.3 0.0.0.0 a 0
net 12.12.5.1 0.0.0.0 a 0
redi eigrp 12 sub

router ei 30
no au
net 150.2.43.1 0.0.0.0
redi os 1 met 1 1 1 1 1

```
R3#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRR, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       N1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, I1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 12.12.5.2 to network 0.0.0.0

     12.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
O E2    12.12.1.0/24 [110/20] via 12.12.5.2, 00:25:21, Serial0/1
C       12.12.5.0/24 is directly connected, Serial0/1
O       12.12.6.0/24 [110/128] via 12.12.5.2, 00:25:21, Serial0/1
O E2    12.12.7.0/24 [110/20] via 12.12.5.2, 00:25:21, Serial0/1
O       12.12.11.1/32 [110/129] via 12.12.5.2, 00:25:21, Serial0/1
C       12.12.33.0/24 is directly connected, Loopback0
O E2    12.12.22.0/24 [110/20] via 12.12.5.2, 00:25:23, Serial0/1
O E2    12.12.44.0/24 [110/20] via 12.12.5.2, 00:25:02, Serial0/1
O       12.12.55.0/24 [110/65] via 12.12.5.2, 00:25:23, Serial0/1
O       12.12.66.0/24 [90/409600] via 150.2.12.254, 00:36:10, FastEthernet0/1
     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O E2    10.8.2.2/32 [110/20] via 12.12.5.2, 00:17:06, Serial0/1
O E2    10.8.8.0/24 [110/20] via 12.12.5.2, 00:17:08, Serial0/1
O E2    10.8.9.0/24 [110/20] via 12.12.5.2, 00:17:08, Serial0/1
     150.2.0.0/24 is subnetted, 1 subnets
C       150.2.43.0 is directly connected, FastEthernet0/1
O*E2 0.0.0.0/0 [110/1] via 12.12.5.2, 00:25:25, Serial0/1
```

1-4 R4

```
int lo0
ip add 12.12.44.4 255.255.255.0

int f0/0
ip add 12.12.2.2 255.255.255.0
no sh

int f0/1
ip add 12.12.1.1 255.255.255.0
no sh

router os 1
router-id 4.4.4.4
net 12.12.44.4 0.0.0.0 a 0
net 12.12.2.2 0.0.0.0 a 0
net 12.12.1.1 0.0.0.0 a 0

ip route 0.0.0.0 0.0.0.0 12.12.1.2
```

```
R4#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, I1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set
     43.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
D EX    43.43.1.0 [170/2560025856] via 150.2.43.1, 00:32:14, FastEthernet0/1
D EX    43.43.5.0 [170/2560025856] via 150.2.43.1,  00:40:16, FastEthernet0/1
D EX    43.43.6.0 [170/2560025856] via 150.2.43.1,  00:40:16, FastEthernet0/1
D EX    43.43.7.0 [170/2560025856] via 150.2.43.1,  00:40:07, FastEthernet0/1
D EX    43.43.11.0 [170/2560025856] via 150.2.43.1, 00:40:09, FastEthernet0/1
D EX    43.43.33.0 [170/2560025856] via 150.2.43.1, 00:40:18, FastEthernet0/1
D EX    43.43.34.0 [170/2560025856] via 150.2.43.1, 00:32:25, FastEthernet0/1
D EX    43.43.44.0 [170/2560025856] via 150.2.43.1, 00:29:16, FastEthernet0/1
D EX    43.43.55.0 [170/2560025856] via 150.2.43.1, 00:29:47, FastEthernet0/1
C       43.43.66.0 is directly connected, Loopback0
     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
D EX    10.8.2.2/32 [170/2560025856] via 150.2.43.1, 00:21:20, FastEthernet0/1
D EX    10.8.8.0/24 [170/2560025856] via 150.2.43.1, 00:21:21, FastEthernet0/1
D EX    10.8.9.0/24 [170/2560025856] via 150.2.43.1, 00:21:21, FastEthernet0/1
     150.2.0.0/24 is subnetted, 1 subnets
C       150.2.43.0 is directly connected, FastEthernet0/1
D*EX 0.0.0.0/0 [170/2560025856] via 150.2.43.1, 00:40:15, FastEthernet0/1
```

1-5 R5

int lo0
ip add 12.12.55.5 255.255.255.0

int f0/0
no sh

int f0/0.202
en dot 202
ip add 12.12.19.1 255.255.255.0

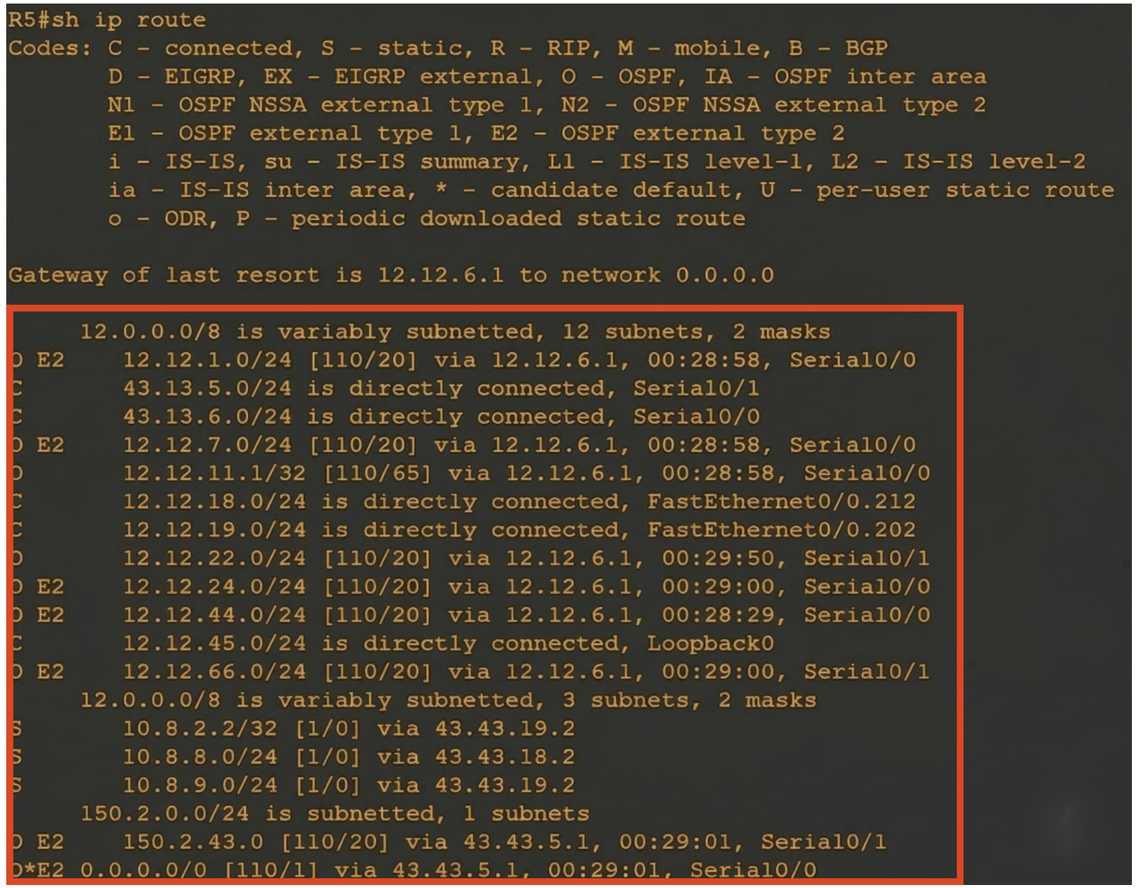
int f0/0.212
en dot 212
ip add 12.12.18.1 255.255.255.0

int s0/0
no sh
ip add 12.12.6.2 255.255.255.0

int s0/1
no sh
ip add 12.12.5.2 255.255.255.0

router os 1
router-id 5.5.5.5
net 12.12.55.5 0.0.0.0 a 0
net 12.12.5.2 0.0.0.0 a 0
net 12.12.6.2 0.0.0.0 a 0
redi static sub

ip route 10.8.8.0 255.255.255.0 12.12.18.2
ip route 10.8.9.0 255.255.255.0 12.12.19.2
ip route 10.8.2.2 255.255.255.255 12.12.19.2



1-6 R6

int lo0
ip add 12.12.66.6 255.255.255.0

int f0/1
no sh
ip add 150.2.43.254 255.255.255.0

router ei 30
no au
net 150.2.43.254 0.0.0.0
net 12.12.66.6 0.0.0.0

```
     43.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
D EX    43.43.1.0/24
           [170/2560025856] via 150.2.43.1, 00:32:14, FastEthernet0/1
D EX    43.43.5.0/24
           [170/2560025856] via 150.2.43.1, 00:40:16, FastEthernet0/1
D EX    43.43.6.0/24
           [170/2560025856] via 150.2.43.1, 00:40:16, FastEthernet0/1
D EX    43.43.7.0/24
           [170/2560025856] via 150.2.43.1, 00:40:07, FastEthernet0/1
D EX    43.43.11.1/32
           [170/2560025856] via 150.2.43.1, 00:40:09, FastEthernet0/1
D EX    43.43.33.0/24
           [170/2560025856] via 150.2.43.1, 00:40:18, FastEthernet0/1
D EX    43.43.34.0/24
           [170/2560025856] via 150.2.43.1, 00:32:25, FastEthernet0/1
D EX    43.43.44.0/24
           [170/2560025856] via 150.2.43.1, 00:29:16, FastEthernet0/1
D EX    43.43.55.0/24
           [170/2560025856] via 150.2.43.1, 00:29:47, FastEthernet0/1
C       43.43.66.0/24 is directly connected, Loopback0
     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
D EX    10.8.2.2/32 [170/2560025856] via 150.2.43.1, 00:21:20, FastEthernet0/1
D EX    10.8.8.0/24 [170/2560025856] via 150.2.43.1, 00:21:21, FastEthernet0/1
D EX    10.8.9.0/24 [170/2560025856] via 150.2.43.1, 00:21:21, FastEthernet0/1
     150.2.0.0/24 is subnetted, 1 subnets
C       150.2.43.0 is directly connected, FastEthernet0/1
D*EX 0.0.0.0/0 [170/2560025856] via 150.2.43.1, 00:40:15, FastEthernet0/1
```

04

방화벽 1 설정

1-1 인터페이스 설정

1-2 Redundant 1

1-3 라우팅

1-4 MPF

1-1 인터페이스 설정

int g0
no sh

int g1
no sh

int g3
no sh

int g1
nameif inside
ip add 12.12.7.1 255.255.255.0

int g0
nameif giga
security-level 100
ip add 12.12.3.2 255.255.255.0

int g3
nameif outside
ip add 143.43.3.2 255.255.255.0

```
interface GigabitEthernet0
 nameif giga
 security-level 100
 ip address 12.12.3.2 255.255.255.0
!
interface GigabitEthernet1
 nameif inside
 security-level 100
 ip address 12.12.7.1 255.255.255.0
!
interface GigabitEthernet2
 shutdown
 no nameif
 no security-level
 no ip address
!
interface GigabitEthernet3
 nameif outside
 security-level 0
 ip address 143.43.3.2 255.255.255.0
```

1-2 Redundant 1

Redundant

시스템이나 네트워크에서 장애 발생 시에도 서비스가 중단되지 않도록 장비나 경로를 이중으로 구성하는 기술

int redundant 1
member-interface g0
member-interface g3

nameif outside
ip add 143.43.3.2 255.255.255.0

```
FW1(config-if)# sh int redundant 1
Interface Redundant1 "outside", is up, line protocol is up
```

```
IP address 143.43.3.2, subnet mask 255.255.255.0
```

```
Redundancy Information:
      Member GigabitEthernet0(Active), GigabitEthernet3
      Last switchover at 11:50:54 UTC Mar 17 2026
```

1-3 라우팅

router os 1
router-id 11.11.11.11
net 143.43.3.2 255.255.255.255 a 0
redi eigrp 12 sub

router ei 12
no au
net 12.12.7.1 255.255.255.255
redi os 1 met 1 1 1 1 1

```
FW1(config)# sh route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRF external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    12.12.1.0 255.255.255.0 [110/20] via 12.12.34.2, 0:46:08, outside
D EX 12.12.5.0 255.255.255.0 [170/2560002816] via 12.12.7.2, 0:49:27, inside
D EX 12.12.6.0 255.255.255.0 [170/2560002816] via 12.12.7.2, 0:49:27, inside
C    12.12.7.0 255.255.255.0 is directly connected, inside
D EX 12.12.11.0 255.255.255.0 [170/2560002816] via 12.12.7.2, 0:49:27, inside
D EX 12.12.33.0 255.255.255.0 [170/2560002816] via 12.12.7.2, 0:46:29, inside
C    12.12.34.0 255.255.255.0 is directly connected, outside
O    12.12.44.0 255.255.255.0 [110/11] via 12.12.34.2, 0:46:08, outside
D EX 12.12.55.0 255.255.255.0 [170/2560002816] via 12.12.7.2, 0:46:39, inside
D EX 12.12.66.0 255.255.255.0 [170/2560002816] via 12.12.7.2, 0:49:27, inside
D EX 10.8.2.2 255.255.255.255 [170/2560002816] via 12.12.7.2, 0:38:12, inside
D EX 10.8.8.0 255.255.255.0 [170/2560002816] via 12.12.7.2, 0:38:13, inside
D EX 10.8.9.0 255.255.255.0 [170/2560002816] via 12.12.7.2, 0:38:13, inside
D EX 150.2.43.0 255.255.255.0 [170/2560002816] via 12.12.7.2, 0:49:27, inside
```

1-4 MPF

MPF

방화벽에서 트래픽을 분류하고, 이에 대한 처리 정책을 정의하여 적용하는 정책 관리 기술

class-map inspection_default
match default-inspection-traffic

```
FW1(config)# sh run policy-map
!
policy-map global_policy
 class inspection_default
  inspect icmp
!
```

policy-map global_policy
class inspection_default
inspect icmp

service-policy global_policy int inside
service-policy global_policy int outside

```
FW1(config)# show service-policy

Interface outside:
  Service-policy: global_policy
    Class-map: inspection_default
      Inspect: icmp, packet 88  drop 0, reset-drop 0

Interface inside:
  Service-policy: global_policy
    Class-map: inspection_default
      Inspect: icmp, packet 20  drop 0, reset-drop 0
```

내부 → 외부 Ping (R1 → R4)

```
R1#ping 12.12.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.12.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/88/152 ms
```

04

방화벽 2 설정

01 Active Key 설정

02 Context 설정

03 ACL

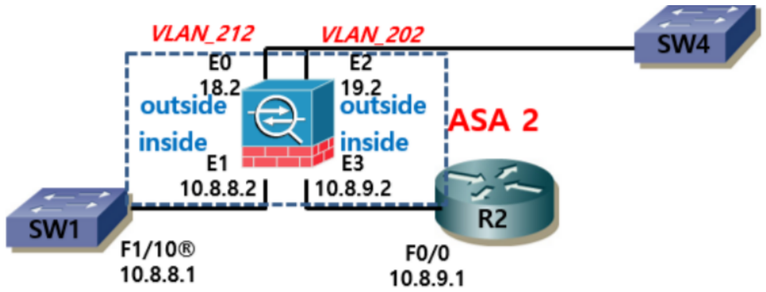
04 라우팅

05 Object NAT

1-1 Active Key 설정

Security Context

하나의 방화벽을 여러 개의 가상 방화벽처럼 나누어 사용하는 기술



1. Active Key 입력

activation-key 0×4a3ec071 0×0d86fbf6 0×7cb1bc48 0×8b48b8b0 0xf317c0b5

2. reload(재부팅)

3. mode multiple

```
FW2(config)# sh mode
Security context mode: multiple
```

1-2 Context 설정

인터페이스 설정

int g0
no sh

int g1
no sh

int g2
no sh

int g3
no sh

admin-context admin
context admin
config-url admin.cfg

context C1
allocate-int g0
allocate-int g1
config-url C1.cfg

context C2
allocate-int g2
allocate-int g3
config-url C2.cfg

```
FW2(config)# sh context
Context Name       Class       Interfaces              URL
*admin             default                             disk0:/admin.cfg
 C1                default     GigabitEthernet0,       disk0:/C1.cfg
                               GigabitEthernet1
 C2                default     GigabitEthernet2,       disk0:/C2.cfg
                               GigabitEthernet3

Total active Security Contexts: 3
```

1-2 Context 설정

changeto context C1
int g1
nameif inside
ip add 10.8.8.2 255.255.255.0

```
FW2/C1(config-if)# sh int inside
Interface GigabitEthernet1 "inside", is up, line protocol is up
        MAC address 0000.ab6f.5501, MTU 1500
        IP address 10.8.8.2, subnet mask 255.255.255.0
  Traffic Statistics for "inside":
        0 packets input, 0 bytes
        1 packets output, 28 bytes
        0 packets dropped
```

int g0
nameif outside
ip add 12.12.18.2 255.255.255.0

```
FW2/C1(config-if)# sh int outside
Interface GigabitEthernet0 "outside", is up, line protocol is up
        MAC address 0000.ab6f.5500, MTU 1500
        IP address 12.12.18.2, subnet mask 255.255.255.0
  Traffic Statistics for "outside":
        0 packets input, 0 bytes
        1 packets output, 28 bytes
        0 packets dropped
```

1-2 Context 설정

changeto context C2
int g3
nameif inside
ip add 10.8.9.2 255.255.255.0

```
interface GigabitEthernet3
 nameif inside
 security-level 100
 ip address 10.8.9.2 255.255.255.0
!
```

int g2
nameif outside
ip add 12.12.19.2 255.255.255.0

```
FW2/C2(config-if)# sh int outside
Interface GigabitEthernet2 "outside", is up, line protocol is up
        MAC address 0000.ab2b.8602, MTU 1500
        IP address 12.12.19.2, subnet mask 255.255.255.0
  Traffic Statistics for "outside":
        0 packets input, 0 bytes
        1 packets output, 28 bytes
        0 packets dropped
```

1-3 ACL

changeto context C1
access-list acl_oi per icmp any any
access-group acl_oi in interface outside

```
FW2/C1(config)# sh access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096)
            alert-interval 300
access-list acl_oi; 1 elements; name hash: 0x4bf52f3b
access-list acl_oi line 1 extended permit icmp any any (hitcnt=0) 0x865e8c90
```

내부 → 외부 Ping (SW1 → R5)

```
L3_SW1#ping 12.12.18.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.12.18.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/80/152 ms
```

외부 → 내부 Ping (R5 → SW1)

```
R5#ping 10.8.8.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.8.8.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/95/220 ms
```

1-3 ACL

changeto context C2
access-list acl_oi per icmp any any
access-group acl_oi in interface outside

```
FW2/C2(config)# sh access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096)
            alert-interval 300
access-list acl_oi; 1 elements; name hash: 0x4bf52f3b
access-list acl_oi line 1 extended permit icmp any any (hitcnt=0) 0x865e8c90
```

내부 → 외부 Ping (R2 → R5)

```
R2#ping 12.12.19.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.12.19.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 36/76/96 ms
```

내부 → 외부 Ping (R5 → R2)

```
R5#ping 10.8.9.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.8.9.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/110/280 ms
```

1-4 라우팅

changeto context C1
route outside 0 0 12.12.18.1
route inside 10.8.0.0 255.255.0.0 10.8.8.1

```
FW2/C1(config)# sh route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 12.12.18.1 to network 0.0.0.0

S     10.8.0.0 255.255.0.0 [1/0] via 10.8.8.1, inside
C     10.8.8.0 255.255.255.0 is directly connected, inside
C     12.12.18.0 255.255.255.0 is directly connected, outside
S*    0.0.0.0 0.0.0.0 [1/0] via 12.12.18.1, outside
```

changeto context C2
route outside 0 0 12.12.18.2
route inside 10.8.2.0 255.255.0.0 10.8.9.1

```
FW2/C2(config)# sh route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 12.12.18.2 to network 0.0.0.0

S     10.8.0.0 255.255.0.0 [1/0] via 10.8.9.1, inside
C     10.8.9.0 255.255.255.0 is directly connected, inside
C     12.12.19.0 255.255.255.0 is directly connected, outside
S*    0.0.0.0 0.0.0.0 [1/0] via 12.12.18.2, outside
```

1-5 Object NAT

C1 Static Object NAT - 내부 IP 주소와 외부 IP 주소를 1:1로 고정 매핑하는 주소 변환 기술

changeto context C1
object network ob_static
host 10.8.8.1
nat (inside,outside) static 12.12.18.3

```
FW2/C1# sh nat

Auto NAT Policies (Section 2)
1 (inside) to (outside) source static ob_static 12.12.18.3
    translate_hits = 0, untranslate_hits = 0
```

C2 Dynamic Object NAT - 내부 IP 주소를 사용 가능한 외부 IP 주소 중 하나로 1:1로 매핑하는 주소 변환 기술

changeto context C2
object network ob_dynamic
subnet 10.8.0.0 255.255.0.0
nat (inside,outside) dynamic interface

```
FW2/C2# sh nat

Auto NAT Policies (Section 2)
1 (inside) to (outside) source dynamic ob_dynamic interface
    translate_hits = 0, untranslate_hits = 0
```

1-5 Object NAT

static nat 설정 전

```
R5#ping 10.8.8.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.8.8.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/96/256 ms
```

R5 → SW1

static nat 설정 후

```
R5#ping 10.8.8.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.8.8.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

R5 → SW1

```
FW2/C1(config)# sh xlate
1 in use, 1 most used
Flags: D - DNS, i - dynamic, r - portmap, s - s
NAT from inside:10.8.8.1 to outside:12.12.18.3
    flags s idle 0:00:38 timeout 0:00:00
```

```
R5#ping 12.12.18.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.12.18.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/68/84 ms
```

SW1 → R5

3조

감사합니다

한준영, 서승덕, 백우진, 양혜원

| Fin |